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International Mathematics Assessments for Schools

2021 ~ 2022 Upper Primary Division 2nd Round Paper

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

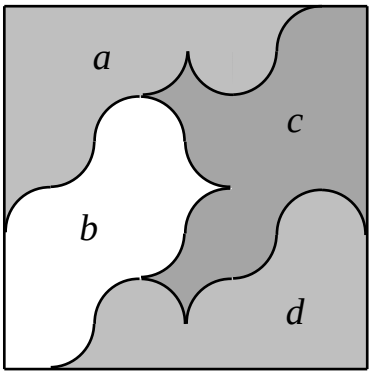
2021 ~ 2022 Upper Primary Division 2nd Round Paper

Questions 1 to 5, 4 marks each

1. Robert arranged his favorite books by writing numbers in the following manner: 1, 2, 3, …. It is known that the sum of all the numbers he wrote is a multiple of 100 and less than 1000. What is the largest number that Robert wrote in a book?
（A）20　（B）22　（C）24　（D）25　（E）28

Answer：_____

2. The diagram below shows a square that is formed using four parts.



Which two parts have the same area?
（A）*a* and *b*　（B）*a* and *c*　（C）*b* and *c*
（D）*b* and *d*　（E）*c* and *d*

Answer：_____

3. In the sequence,

1, 2, 4, 8, 6, …

Starting from the second term, each term is formed by multiplying the previous term by 2 and then taking the remainder when divided by 10. What is the 2022nd term?
（A）1　（B）2　（C）4　（D）6　（E）8

Answer：_____

4. How many different four-digit odd numbers can be formed using the digits 0, 1, 2, 3 and 4 if one can repeat digits? (Note that 0 is not allowed to be the leading digit.)
（A）48　（B）200　（C）250　（D）500　（E）625

Answer：_____

5. Divide the numbers 1, 2, …, 12 into 4 groups, such that each group contains three numbers and the sum of two of them is equal to the third. What is the smallest product of those four sums?

（A）8748　（B）8640　（C）8316　（D）7920　（E）7260

Answer：_____

Questions 6 to 13, 5 marks each

6. There are two stores that sells the same item. Store A sells the item at list price and sold them all and got a total of $7200. Store B sells the same item at a 20% discount of the list price and sold 15 more pieces than store A. If it is known that the two stores have the same income, how many pieces of the item did store A sell?

Answer：_____ pieces

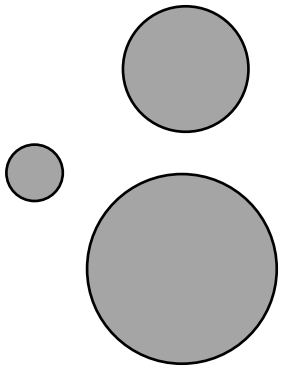
7. Larry owns a cage that can hold a total of 15 peacocks and parrots. If the food box inside the cage is completely filled, the birds can consume all of it in 6 days. Later on, one parrot was added to the cage. The food box was full at the start and $\frac{1}{16}$ of the feeds is added every day. After exactly 9 days, the feed box was completely empty. It is known that the peacock eats twice as much feeds as the parrot does every day. Moreover, both kinds of birds eat a fixed amount of feeds every day. How many peacocks are there inside the cage?

Answer：_____ peacocks

8. The numbers from 1, 2, 3, …, 100 are written on a board. Peter first calculates the sum of these numbers. Then Mary takes the sum calculated by Peter and deducts from it the sum of all digits written on the board. What result does Mary get?

Answer：_____

9. How many different ways can you draw a fourth circle on the plane that contains three disjoint circles, as shown in the diagram, such that all three circles touch the circumference of the fourth one?

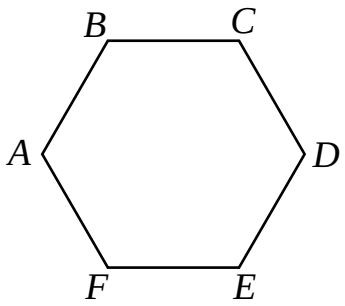


Answer : _____ ways

10. There are four different positive integers such that their product is seven times of the product of each number subtracted by 3. What is the minimum value of the product of these four integers?

Answer : _____

11. How many ways can we draw three non-intersecting diagonals inside regular hexagon *ABCDEF* such that the figure will be divided into 4 triangles (not necessarily all triangles have to be of the same size) ?

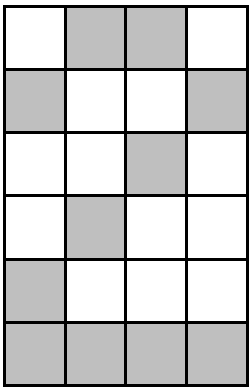


Answer : _____ ways

12. In a line, there is at least one girl and at least one boy. Two children with exactly 4 or 7 other children between them are of the same gender. What is the maximum number of children in this line?

Answer : _____ children

13. Some unit squares on a 6×4 board are coloured gray, as shown in the diagram below. Find the number of ways in choosing four different gray unit squares such that any two unit squares that are chosen are located on different rows and different columns.



Answer： _____ ways

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

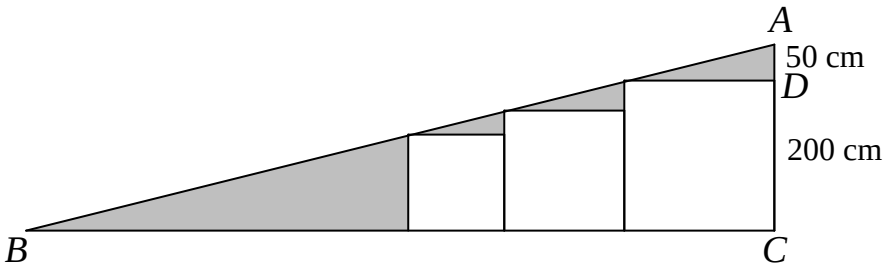
14. A continuous look-and-say sequence is formed this way: Start with three 1's; read the terms as "three 1's" and write down what you read as "31" at the end of the sequence; read the next group of unread identical numbers as "one 3" and write "13" at the end; continue this way to get a infinite sequence 1, 1, 1, 3, 1, 1, 3, 2, 1, ⋯.
    (a) What is the largest possible digit you could write down? Why? (5 points)
    (b) What is the 20th term of this sequence? Why? (5 points)
    (c) Will the number "333" appear in the sequence? Why or why not? (10 points)

Answer： (a) The largest possible digit is _____ . The reason is:

(b) The 20th term of this sequence is _____ . The reason is:

(c) ☐ Yes. ☐ No. The reason is:

15. In the diagram below, ABC is a right triangle where three squares of different sizes are drawn inside it. If $AD = 50$ cm and $CD = 200$ cm, then what is the area, in cm^2, of the shaded region?



Answer : _____ cm^2